

Mathematics for Machine Learning

Vazgen Mikayelyan



Definition

The trace of a square matrix $A \in \mathbb{R}^{n \times n}$ is given by

$$\text{tr}(A) := \sum_{i=1}^n a_{ii}$$

in other words, the trace is the sum of the diagonal elements of A .

Definition

The trace of a square matrix $A \in \mathbb{R}^{n \times n}$ is given by

$$\text{tr}(A) := \sum_{i=1}^n a_{ii}$$

in other words, the trace is the sum of the diagonal elements of A .

Remark

For $A, B \in \mathbb{R}^{n \times n}$ the trace satisfies the following properties:

1. $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$
2. $\text{tr}(\alpha A) = \alpha \text{tr}(A)$
3. $\text{tr}(I_n) = n$
4. $\text{tr}(AB) = \text{tr}(BA)$

Corollary

- In particular, for two vectors $x, y \in \mathbb{R}^n$

$$\text{tr}(\mathbf{x}\mathbf{y}^T) = \text{tr}(\mathbf{y}^T\mathbf{x}) = \mathbf{y}^T\mathbf{x} \in \mathbb{R}.$$

- If $A \sim B$ (i.e. $\exists Q$ s.t. $A = QBQ^{-1}$), then $\text{tr}A = \text{tr}B$
- All transformation matrices A_T of a linear mapping $T : V \rightarrow V$ have the same trace.

Corollary

- In particular, for two vectors $x, y \in \mathbb{R}^n$

$$\text{tr}(\mathbf{x}\mathbf{y}^T) = \text{tr}(\mathbf{y}^T\mathbf{x}) = \mathbf{y}^T\mathbf{x} \in \mathbb{R}.$$

- If $A \sim B$ (i.e. $\exists Q$ s.t. $A = QBQ^{-1}$), then $\text{tr} A = \text{tr} B$
- All transformation matrices A_T of a linear mapping $T : V \rightarrow V$ have the same trace.

Definition

For $\lambda \in \mathbb{R}$ and a square matrix $A \in \mathbb{R}^{n \times n}$

$$p_A(\lambda) = \det(A - \lambda I) = c_0 + c_1\lambda + c_2\lambda^2 + \dots + c_{n-1}\lambda^{n-1} + (-1)^n\lambda^n$$

$c_0, \dots, c_{n-1} \in \mathbb{R}$, is the **characteristic polynomial** of A . In particular,

$$c_0 = \det(A), \quad c_{n-1} = (-1)^{n-1}\text{tr}(A).$$

Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

Definition

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. Then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A and a nonzero $\mathbf{x} \in \mathbb{R}^n$ is the corresponding **eigenvector** of A if

$$A\mathbf{x} = \lambda\mathbf{x}.$$

Eigenvalues and Eigenvectors

Definition

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. Then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A and a nonzero $\mathbf{x} \in \mathbb{R}^n$ is the corresponding **eigenvector** of A if

$$A\mathbf{x} = \lambda\mathbf{x}.$$

Remark

If $\mathbf{x}$ is an eigenvector of A , then so is $c\mathbf{x}$, for any nonzero $c \in \mathbb{R}$.

Eigenvalues and Eigenvectors

Definition

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. Then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A and a nonzero $\mathbf{x} \in \mathbb{R}^n$ is the corresponding **eigenvector** of A if

$$A\mathbf{x} = \lambda\mathbf{x}.$$

Remark

If $\mathbf{x}$ is an eigenvector of A , then so is $c\mathbf{x}$, for any nonzero $c \in \mathbb{R}$.

Theorem

$\lambda \in \mathbb{R}$ is eigenvalue of $A \in \mathbb{R}^{n \times n}$ if and only if λ is a root of the characteristic polynomial $p_A(\lambda)$ of A .

Eigenvalues and Eigenvectors

Definition

*For $A \in \mathbb{R}^{n \times n}$ the set of all eigenvectors of A associated with an eigenvalue λ together with the zero vector is called **eigenspace of A** with respect to λ and is denoted by E_λ . The set of all eigenvalues of A is called **spectrum** of A .*

Eigenvalues and Eigenvectors

Definition

For $A \in \mathbb{R}^{n \times n}$ the set of all eigenvectors of A associated with an eigenvalue λ together with the zero vector is called **eigenspace of A** with respect to λ and is denoted by E_λ . The set of all eigenvalues of A is called **spectrum** of A .

Example

Find the eigenvalues and eigenvectors of a 2×2 matrix $A = \begin{bmatrix} 5 & 8 \\ 2 & 5 \end{bmatrix}$

Eigenvalues and Eigenvectors

Definition

For $A \in \mathbb{R}^{n \times n}$ the set of all eigenvectors of A associated with an eigenvalue λ together with the zero vector is called **eigenspace of A** with respect to λ and is denoted by E_λ . The set of all eigenvalues of A is called **spectrum** of A .

Example

Find the eigenvalues and eigenvectors of a 2×2 matrix $A = \begin{bmatrix} 5 & 8 \\ 2 & 5 \end{bmatrix}$

Proposition

- If $A \sim B$, then eigenvalues of A and B are the same.
- A linear mapping T has eigenvalues that are independent of the choice of basis of its transformation matrix.

Proposition

- *A matrix A and its transpose A^T have the same eigenvalues, but not necessarily the same eigenvectors.*
- *If S is a symmetric matrix, then all its eigenvalues are real.*
- *The eigenvalues of a symmetric positive definite matrix are positive (and real).*
- *The eigenvectors of symmetric matrices (corresponding to different eigenvalues) are always orthogonal to each other.*

Proposition

- *A matrix A and its transpose A^T have the same eigenvalues, but not necessarily the same eigenvectors.*
- *If S is a symmetric matrix, then all its eigenvalues are real.*
- *The eigenvalues of a symmetric positive definite matrix are positive (and real).*
- *The eigenvectors of symmetric matrices (corresponding to different eigenvalues) are always orthogonal to each other.*

Theorem

For any matrix $A \in \mathbb{R}^{m \times n}$ the matrices $A^T A$ and AA^T are symmetric and positive definite.

Proposition

- *A matrix A and its transpose A^T have the same eigenvalues, but not necessarily the same eigenvectors.*
- *If S is a symmetric matrix, then all its eigenvalues are real.*
- *The eigenvalues of a symmetric positive definite matrix are positive (and real).*
- *The eigenvectors of symmetric matrices (corresponding to different eigenvalues) are always orthogonal to each other.*

Theorem

For any matrix $A \in \mathbb{R}^{m \times n}$ the matrices $A^T A$ and AA^T are symmetric and positive definite.

Theorem

Any symmetric matrix $A = A^T \in \mathbb{R}^{n \times n}$ has n independent eigenvectors that form an orthogonal basis for $\mathbb{R}^n$.

Definition

*Let a square matrix A has an eigenvalue λ . The **algebraic multiplicity** of λ is the number of times the root appears in the characteristic polynomial.*

Definition

Let a square matrix A has an eigenvalue λ . The **algebraic multiplicity** of λ is the number of times the root appears in the characteristic polynomial.

Definition

Let a square matrix A has an eigenvalue λ . The **geometric multiplicity** of λ is the total number of linearly independent eigenvectors associated with λ . In other words it is the dimension of the eigenspace E_λ spanned by the eigenvectors associated with λ .

Definition

Let a square matrix A has an eigenvalue λ . The **algebraic multiplicity** of λ is the number of times the root appears in the characteristic polynomial.

Definition

Let a square matrix A has an eigenvalue λ . The **geometric multiplicity** of λ is the total number of linearly independent eigenvectors associated with λ . In other words it is the dimension of the eigenspace E_λ spanned by the eigenvectors associated with λ .

Example

Find the geometric and algebraic multiplicity of the eigenvalue(s) of the

matrix $A = \begin{bmatrix} 3 & 0 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Theorem

The determinant of a matrix $A \in \mathbb{R}^{n \times n}$ is the product of its eigenvalues, i.e.,

$$\det(A) = \prod_{i=1}^n \lambda_i$$

where λ_i are (possibly repeated) eigenvalues of A .

Theorem

The determinant of a matrix $A \in \mathbb{R}^{n \times n}$ is the product of its eigenvalues, i.e.,

$$\det(A) = \prod_{i=1}^n \lambda_i$$

where λ_i are (possibly repeated) eigenvalues of A .

Theorem

The trace of a matrix $A \in \mathbb{R}^{n \times n}$ is the sum of its eigenvalues, i.e.,

$$\text{tr}(A) = \sum_{i=1}^n \lambda_i$$

where λ_i are (possibly repeated) eigenvalues of A .

Cholesky Decomposition

Cholesky Decomposition

A symmetric positive definite matrix A can be factorized into a product $A = LL^T$, where L is a lower triangular matrix with positive diagonal elements:

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & \dots & 0 \\ \vdots & \ddots & \vdots \\ l_{n1} & \dots & l_{nn} \end{bmatrix} \begin{bmatrix} l_{11} & \dots & l_{n1} \\ \vdots & \ddots & \vdots \\ 0 & \dots & l_{nn} \end{bmatrix}$$

and L is called the Cholesky factor of A .

Cholesky Decomposition

Cholesky Decomposition

A symmetric positive definite matrix A can be factorized into a product $A = LL^T$, where L is a lower triangular matrix with positive diagonal elements:

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} l_{11} & \dots & 0 \\ \vdots & \ddots & \vdots \\ l_{n1} & \dots & l_{nn} \end{bmatrix} \begin{bmatrix} l_{11} & \dots & l_{n1} \\ \vdots & \ddots & \vdots \\ 0 & \dots & l_{nn} \end{bmatrix}$$

and L is called the Cholesky factor of A .

Example

Find Cholesky decomposition of a general 2×2 symmetric matrix

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{bmatrix}.$$

Applications of Cholesky decomposition

Linear equations. The Cholesky decomposition is mainly used for the numerical solution of linear equations $\mathbf{Ax} = \mathbf{b}$. If A is symmetric and positive definite, then we can solve $\mathbf{Ax} = \mathbf{b}$ by first computing the Cholesky decomposition $\mathbf{A} = \mathbf{LL}^T$, then solving $\mathbf{Ly} = \mathbf{b}$ for y by forward substitution, and finally solving $\mathbf{L}^T\mathbf{x} = \mathbf{y}$ for x by back substitution.

Corollary

If $A = LL^T$, then

$$\det A =$$

Corollary

If $A = LL^T$, then

$$\det A = (\det L)^2 =$$

Corollary

If $A = LL^T$, then

$$\det A = (\det L)^2 = (l_{11} \cdot \dots \cdot l_{nn})^2.$$

Moreover

$$A^{-1} = (L^{-1})^T (L^{-1}).$$

Eigendecomposition and Diagonalization

Definition

A matrix $A \in \mathbb{R}^{n \times n}$ is **diagonalizable** if it is similar to a diagonal matrix, i.e. there exists a matrix $P \in \mathbb{R}^{n \times n}$ so that

$$D = P^{-1}AP,$$

(equivalently $AP = PD$ or $A = PDP^{-1}$) where $D = \begin{bmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{bmatrix}$

Proposition

If $D \in \mathbb{R}^{n \times n}$ is a diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_n$, then

$$AP = PD$$

for invertible matrix P if and only if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A and the $\mathbf{p}_i$ are the corresponding eigenvectors of A , where $P = [\mathbf{p}_1 \dots \mathbf{p}_n]$.